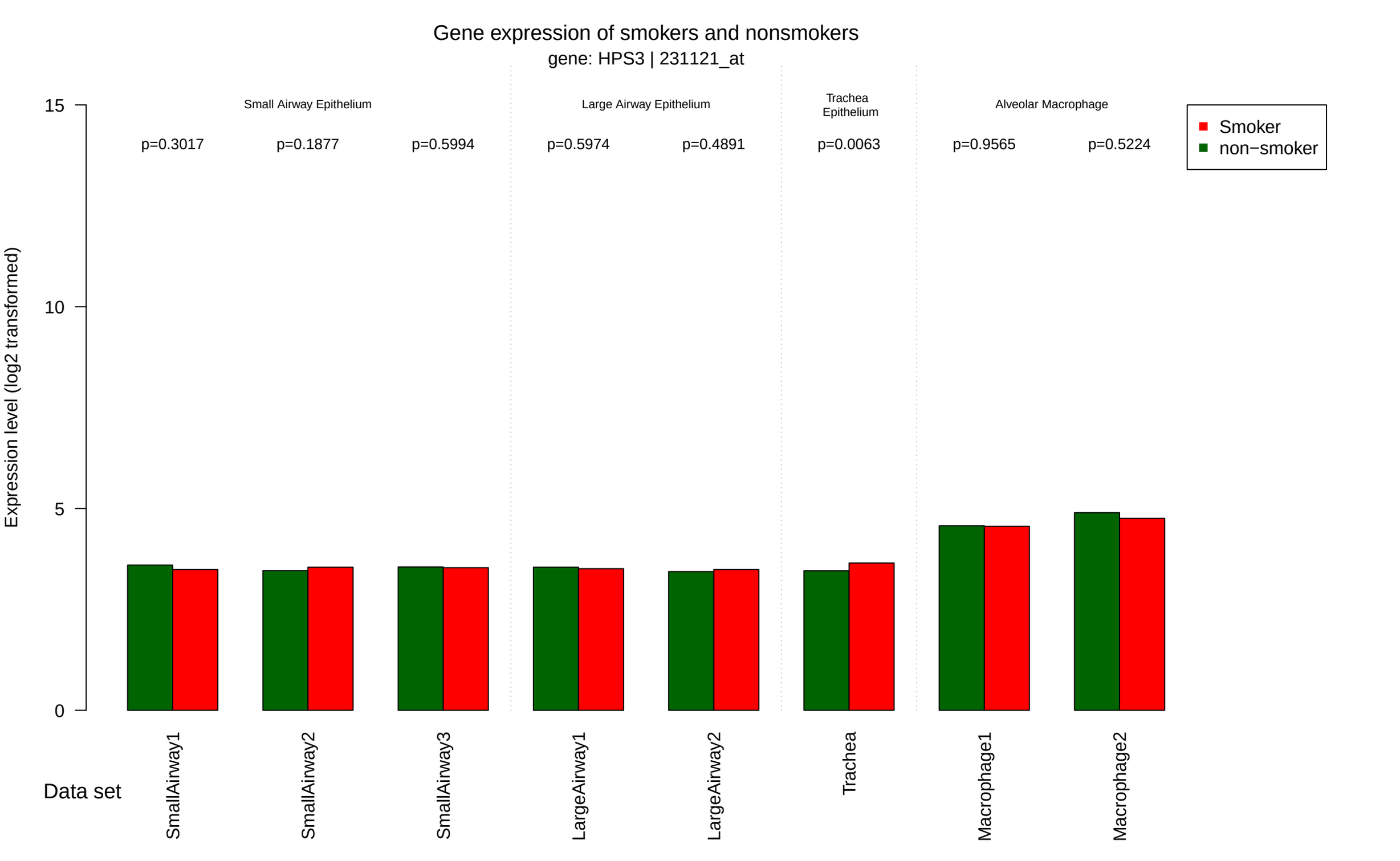
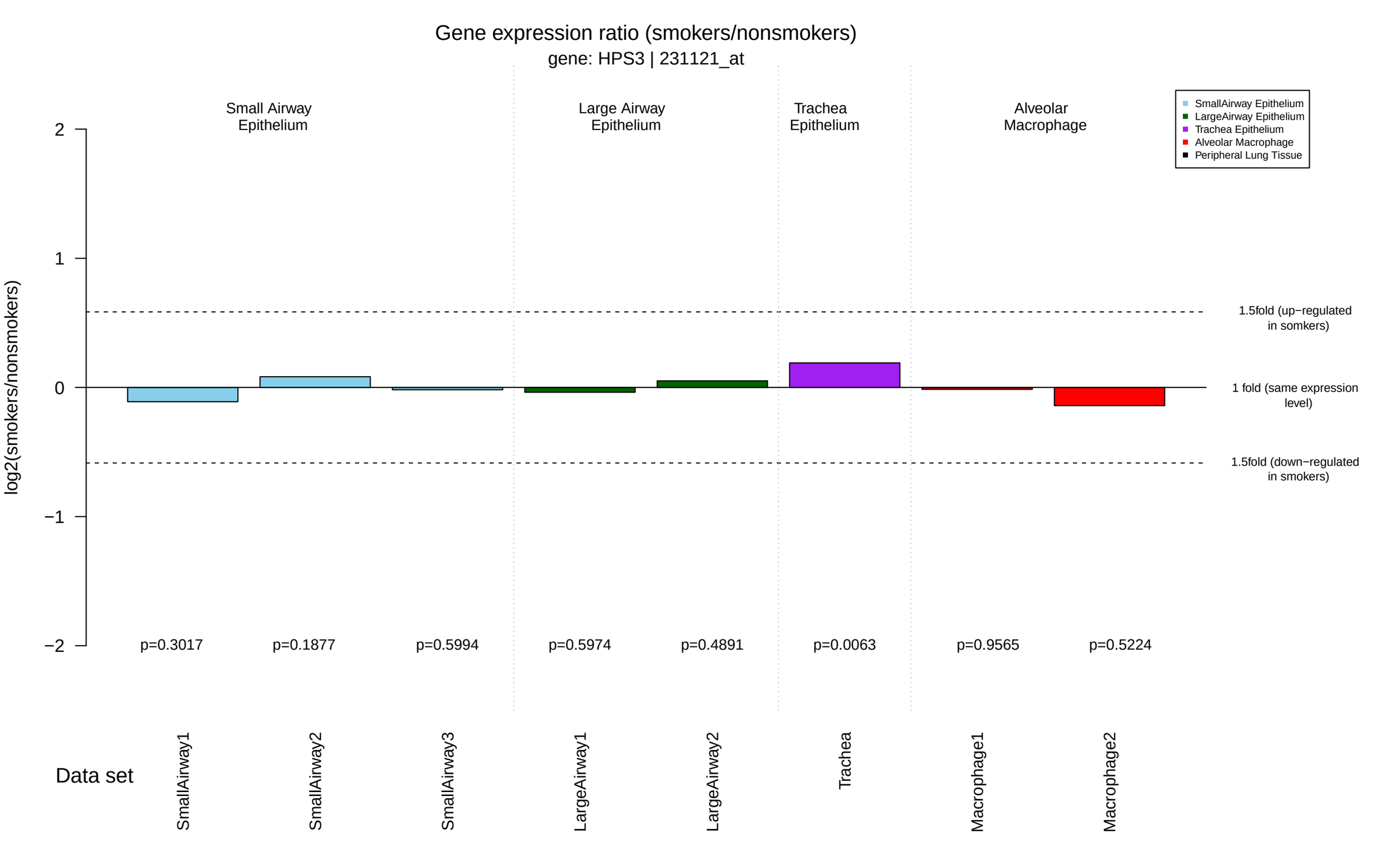
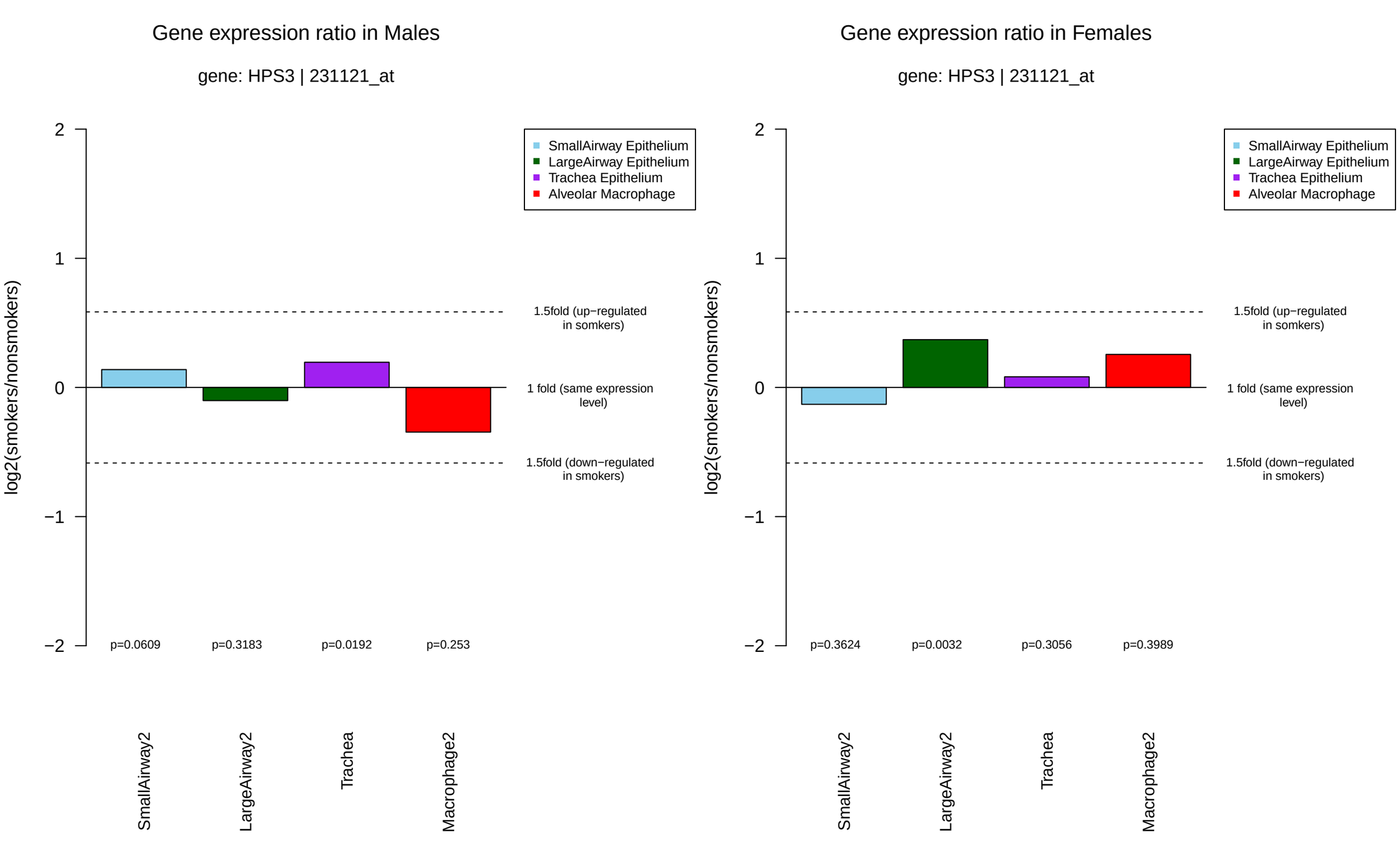
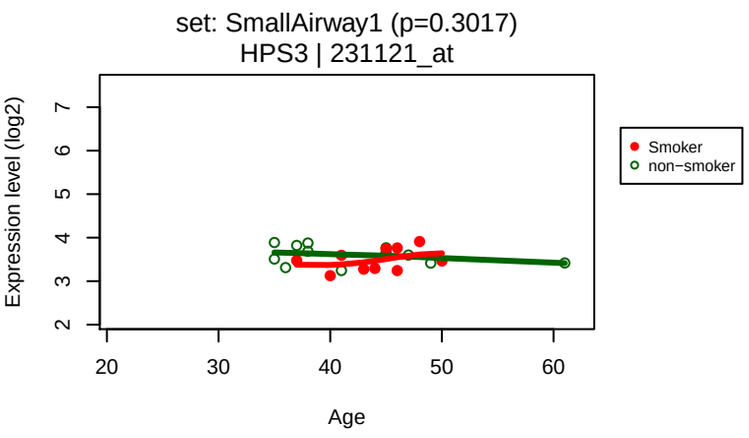
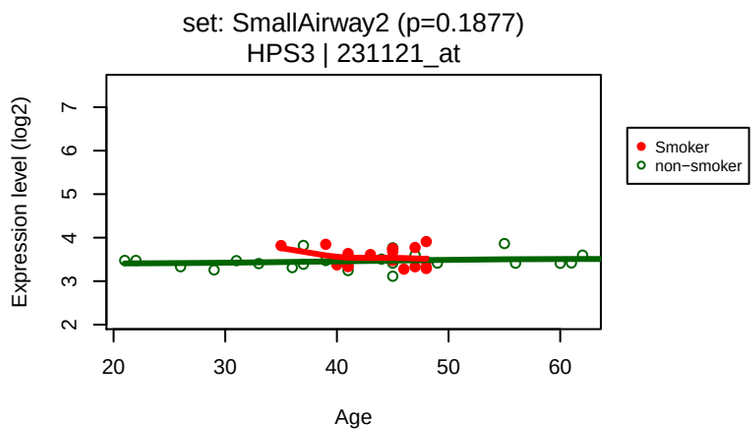
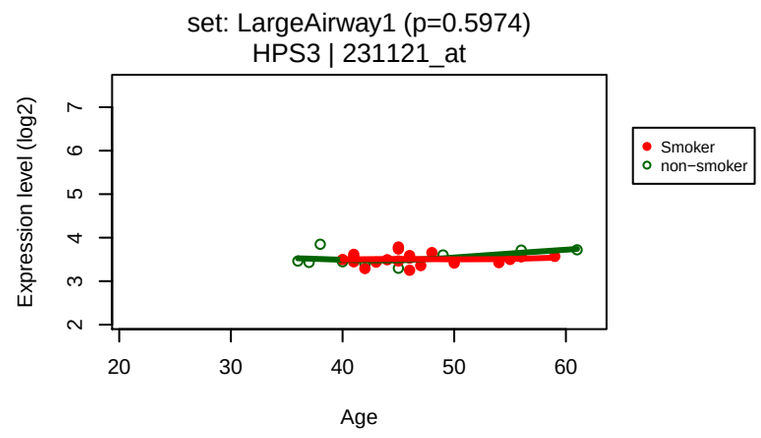

Gene expression of smokers and nonsmokers
gene: HPS3 | 231121_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3017
p=0.1877
p=0.5994
p=0.5974
p=0.4891
p=0.0063
p=0.9565
p=0.5224
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

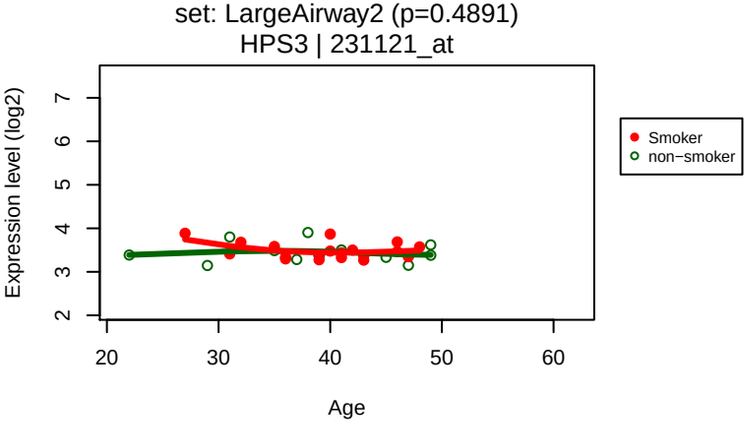

Gene expression ratio (smokers/nonsmokers)
gene: HPS3 | 231121_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3017
p=0.1877
p=0.5994
p=0.5974
p=0.4891
p=0.0063
p=0.9565
p=0.5224
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

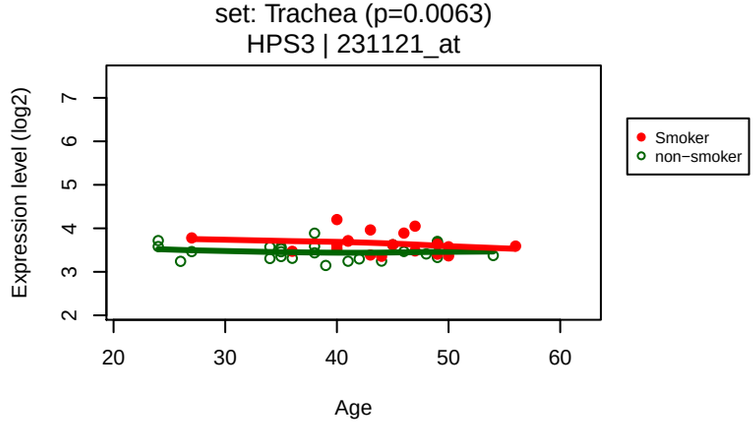

Gene expression ratio in Males
gene: HPS3 | 231121_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0609
p=0.3183
p=0.0192
p=0.253
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HPS3 | 231121_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3624
p=0.0032
p=0.3056
p=0.3989
SmallAirway2
LargeAirway2
Trachea
Macrophage2

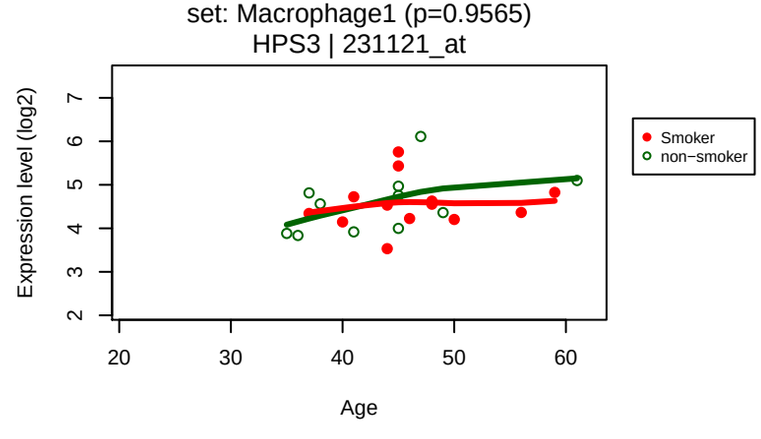

set: SmallAirway1 (p=0.3017)
HPS3 | 231121_at
Expression level (log2)
Age
Smoker
non−smoker

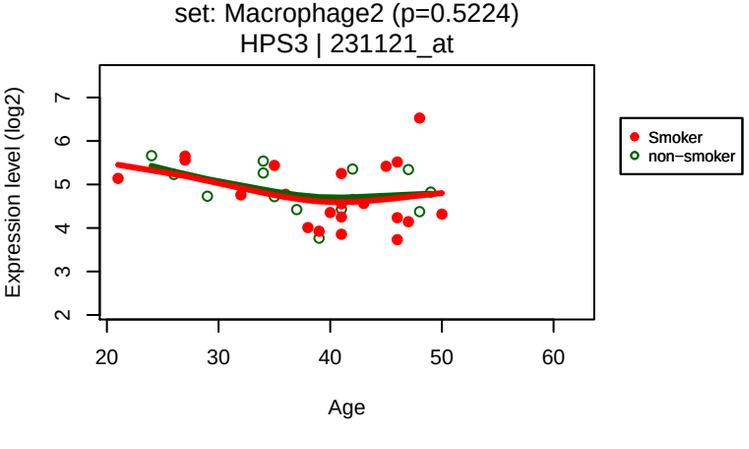

set: SmallAirway2 (p=0.1877)
HPS3 | 231121_at
Expression level (log2)
Age
Smoker
non−smoker

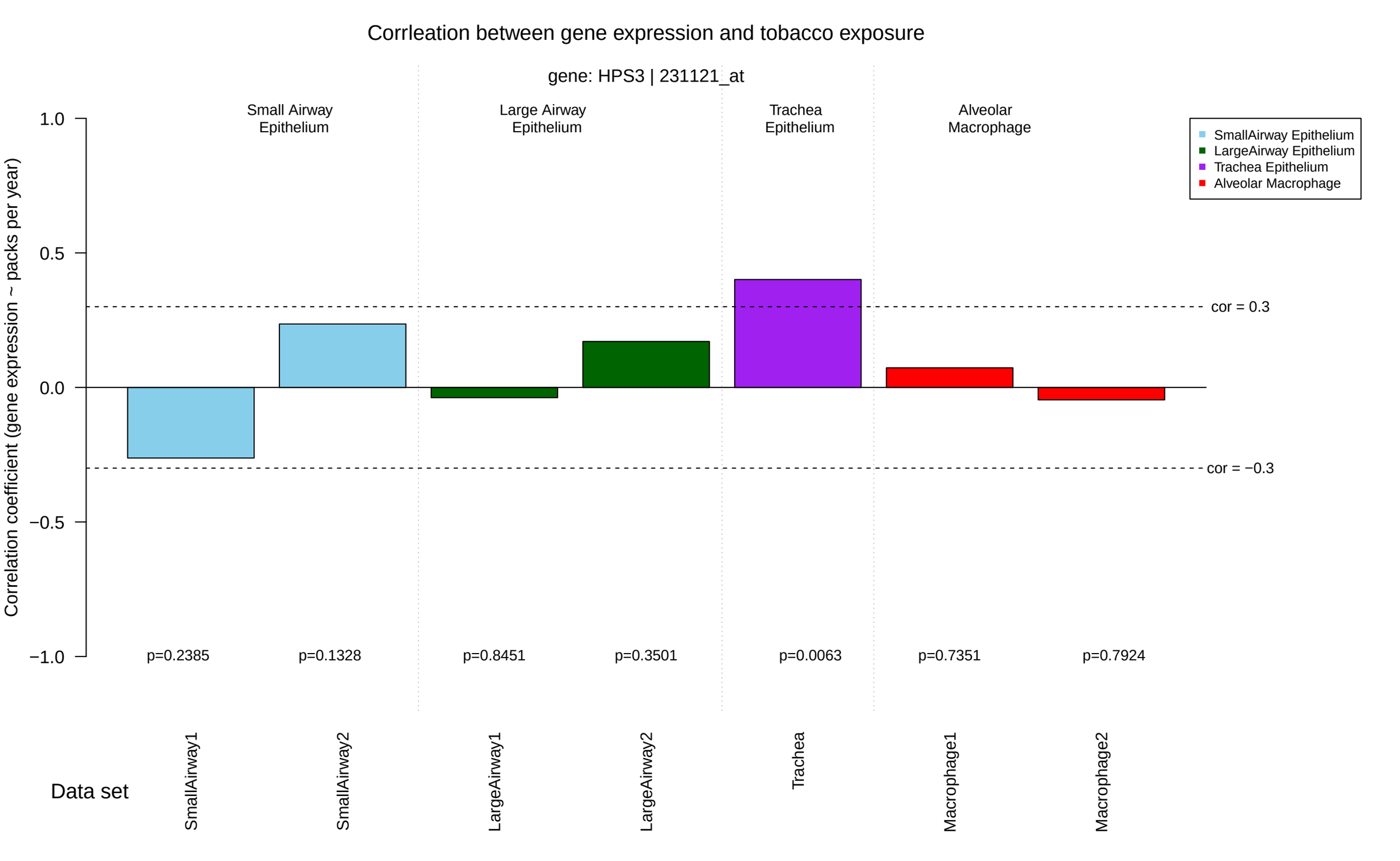

set: LargeAirway1 (p=0.5974)
HPS3 | 231121_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4891)
HPS3 | 231121_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0063)
HPS3 | 231121_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9565)
HPS3 | 231121_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5224)
HPS3 | 231121_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HPS3 | 231121_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2385
p=0.1328
p=0.8451
p=0.3501
p=0.0063
p=0.7351
p=0.7924
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2